1. Project Introduction

Maganor AI: Exploring the Challenges of Detecting Harmful Language Online

The increasing use of social media and digital communication platforms has created new opportunities for people to express themselves, exchange ideas, and connect with others. However, these platforms have also become spaces where individuals experience online harassment, insults, targeted abuse, and gender-based hostility.

One challenge in addressing harmful online communication is that language is complex. A comment may contain offensive words without necessarily being harmful, while another may communicate hostility through sarcasm, indirect expressions, or context-dependent language. This makes automated content moderation particularly challenging.

Maganor AI is an exploratory artificial intelligence project that investigates how machine learning and natural language processing (NLP) can help identify potentially harmful language in online comments.

The project begins with a traditional machine learning approach using TF-IDF and Logistic Regression as a baseline model. It then explores a transformer-based approach using DistilBERT to investigate whether contextual language representations can improve classification performance.

Beyond evaluating model performance, the project examines classification errors, particularly false positives and false negatives, to understand situations in which automated systems incorrectly interpret human language.

The ultimate goal is not simply to build a model that labels comments as toxic or non-toxic, but to investigate the limitations of automated language classification and the importance of context, interpretation, and responsible AI in online safety.

Project Scope

- Explore and analyze an existing online-comment toxicity dataset.
- Develop a baseline text-classification model using TF-IDF and Logistic Regression.
- Fine-tune a pretrained DistilBERT model for toxicity classification.
- Compare the performance of the two approaches.
- Investigate false positives, false negatives, and challenging language examples.
- Discuss dataset limitations, contextual understanding, and responsible AI considerations.

Research Question

How effectively can pretrained language models identify toxic and potentially harmful online comments, and what limitations do they demonstrate when interpreting context, sarcasm, indirect hostility, and gender-based harassment?

2.2 Dataset Structure

The dataset contains the following columns:

Column| Description "id"| Unique identifier for each comment "comment_text"| Original text of the comment "toxic"| Indicates whether the comment is classified as toxic "severe_toxic"| Indicates severe toxic language "obscene"| Indicates obscene language "threat"| Indicates threatening language "insult"| Indicates insulting language "identity_hate"| Indicates identity-based hateful language

Each of the six classification labels is binary:

- "1": The comment has been assigned that label.
- "0": The comment has not been assigned that label.

Importantly, the labels are not mutually exclusive. A single comment can receive multiple labels.

2.3 Dataset Size and Label Distribution

The dataset contains 159,571 comments. Its distribution across the six labels is shown below.

Classification label| Number of comments Toxic| 15,294 Severe toxic| 1,595 Obscene| 8,449 Threat| 478 Insult| 7,877 Identity hate| 1,405

These counts represent the number of comments assigned each individual label. Since labels can overlap, their counts should not be added together to calculate the total number of harmful comments.

2.4 Class Imbalance

The primary classification task in this project uses the "toxic" column to distinguish between toxic and non-toxic comments.

The distribution is:

Class| Number of comments Non-toxic (0)| 144,277 Toxic (1)| 15,294 Total| 159,571

Approximately 90.42% of comments are labeled non-toxic, while 9.58% are labeled toxic.

This indicates a significant class imbalance, meaning that the model encounters considerably more non-toxic examples than toxic examples during training.

Class imbalance is an important consideration because a model may achieve high accuracy simply by favoring the majority class while failing to identify harmful comments.

To address this challenge in the baseline experiment, Logistic Regression was trained using balanced class weights.

2.5 Dataset Relevance to the Research

The Jigsaw dataset is relevant to Maganor AI because it provides labeled examples of several forms of harmful language, allowing the project to investigate automated toxicity classification.

It also provides an opportunity to examine the limitations of automated classification, particularly when comments contain ambiguous expressions, insults, sarcasm, or language that depends on conversational context.

The six available labels allow the project to explore different dimensions of harmful language rather than treating every potentially harmful comment as identical.

2.6 Dataset Limitations

Despite its usefulness, the dataset has several limitations:

1. Language limitation: The dataset consists of English-language comments and does not directly support the classification of Amharic, Afaan Oromo, or other Ethiopian languages.
2. Platform limitation: The comments originate from Wikipedia discussions and may not represent communication patterns on platforms such as TikTok, Facebook, or Instagram.
3. Context limitation: Individual comments may lack the surrounding conversation needed to interpret sarcasm, intention, or indirect hostility.
4. Annotation limitation: Labels represent the dataset's annotations and may not capture every possible interpretation of a comment.
5. Scope limitation: The dataset does not contain a dedicated label for gender-based harassment or suicide risk.

These limitations are important when interpreting model performance and considering the responsible use of AI in online safety.

2.7 Summary

The Jigsaw Toxic Comment Classification dataset provides the foundation for the Maganor AI experiments. Its 159,571 labeled comments allow the development and evaluation of traditional machine learning and transformer-based approaches.

However, model performance on this dataset should not be interpreted as proof that a system can reliably identify all forms of online harassment. Understanding context, annotation limitations, and language diversity remains an important part of the research.

3. Data Exploration and Understanding

Data exploration is an important stage of the research process. Before developing machine learning models, it is necessary to understand the structure, quality, distribution, and characteristics of the dataset.

This stage examines the Jigsaw Toxic Comment Classification dataset to identify patterns, understand the distribution of toxicity labels, investigate class imbalance, and explore the characteristics of the comments.

The findings from this stage help inform preprocessing decisions and model development.

3.1 Initial Dataset Inspection

The first step in data exploration was to examine the dataset's structure and understand the information available for analysis.

The dataset contains 159,571 rows and 8 columns. Each row represents an individual Wikipedia comment, while the columns contain the comment identifier, original text, and six toxicity-related classification labels.

Initial inspection helps establish the dataset's dimensions, identify available variables, and understand how the information is organized before further analysis.

3.2 Data Quality Assessment

Before proceeding with data analysis and model development, the dataset was examined for missing values and duplicate records.

This step is important because missing or duplicated information can affect the reliability of data analysis and model performance.

The assessment focuses on identifying incomplete comments, missing classification labels, and repeated records.

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3.2.1 Missing Values Assessment

The dataset was examined for missing values across all eight columns. The analysis found zero missing values in every column, including the comment text and all six classification labels.

This indicates that every record contains the information required for the planned analysis. Therefore, no missing-value imputation or removal was necessary at this stage.

3.2.2 Duplicate Records Assessment

The dataset was checked for duplicate rows and repeated comment texts.

The analysis identified zero duplicate rows and zero duplicate comment texts. This suggests that the dataset does not contain exact repeated records or identical comments that could unnecessarily influence the analysis.

3.2.3 Empty Comment Assessment

The comment text column was also examined for empty or whitespace-only entries. No empty comments were identified.

This confirms that all 159,571 records contain text that can be used for subsequent natural language processing tasks.

3.2.4 Research Observation

The initial data quality assessment indicates that the dataset is structurally complete, with no detected missing values, duplicate records, or empty comments.

As a result, there was no immediate need for data cleaning through missing-value removal or duplicate elimination. However, further preprocessing and text analysis are still necessary because complete data does not automatically mean that the language itself is consistent, unambiguous, or easy for machine learning models to interpret.

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3.3 Toxicity Label Distribution

The dataset contains six toxicity-related classification labels. To understand the distribution of harmful language categories, t

The results show that the categories are unevenly distributed.

Toxicity category| Number of comments
Toxic| 15,294
Obscene| 8,449
Insult| 7,877
Severe toxic| 1,595
Identity hate| 1,405
Threat| 478

Research Observations

1. Toxic comments: General toxicity has the highest number of labeled examples, with 15,294 comments.

2. Obscene and insulting language: These categories contain 8,449 and 7,877 comments, respectively, making them relatively commo

3. Severe toxicity and identity hate: These categories have considerably fewer examples, with 1,595 and 1,405 comments.

4. Threats: The threat category has the smallest representation, with only 478 comments.

Interpretation

The distribution indicates that the dataset is imbalanced across toxicity categories. In particular, threatening language and id

This imbalance may make it more difficult for machine learning models to learn patterns associated with less frequently represen

Another important consideration is that the six labels are not mutually exclusive. A comment may belong to more than one categor
```

```
3.4 Distribution of Multiple Toxicity Labels

The dataset contains six toxicity-related labels, and individual comments may be assigned more than one label.

To investigate this overlap, the number of toxicity labels assigned to each comment was calculated. This analysis helps determin

Understanding the distribution of multiple labels is important because harmful language is not always limited to a single catego
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Research Observations
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The analysis revealed that most comments (143,346) were assigned none of the six toxicity labels.

Among the labeled comments, 6,360 received one label, while others received multiple labels. A smaller number of comments were a

Interestingly, 31 comments received all six toxicity labels, demonstrating that different forms of harmful language can occur to

Interpretation

These findings demonstrate that harmful language can involve overlapping categories rather than a single isolated characteristic

The distribution also highlights the importance of understanding the dataset's labeling structure when developing classification

Since this project initially focuses on binary toxicity classification using the "toxic" column, the multiple-label analysis pro
```

3.5 Comment Length Analysis

The length of a text can influence how natural language processing models process and interpret written content.

In this analysis, the length of each comment is examined to understand the variation in text size across the dataset. This includes identifying the average comment length, median length, minimum and maximum lengths, and the overall distribution.

Understanding comment length is particularly important when working with transformer-based models such as DistilBERT, which process text using a limited number of tokens per input sequence.

This analysis helps inform decisions about tokenization and maximum sequence length during model development.

3.6 Text Preprocessing and Preparation

Text preprocessing is an important stage in natural language processing because raw text often contains variations in capitalization, punctuation, spacing, and formatting.

In this project, preprocessing is approached carefully to preserve linguistic information that may influence the interpretation of harmful language.

The initial preparation includes examining the original comments, creating a lowercase text representation for the baseline experiment, and retaining the original text for further contextual analysis.

Special attention is given to preserving punctuation, negation, and other linguistic features that may be important when distinguishing harmful expressions from ordinary communication.

3.7 Tokenization and Text Representation

Computers cannot directly understand human language in the same way people do. Natural language processing therefore involves converting written text into representations that machine learning algorithms can process.

Tokenization is the process of dividing text into smaller units called tokens. Depending on the method, these units may be words, subwords, or individual characters.

In this project, tokenization is explored to understand how written comments can be transformed into smaller units for computational analysis.

Different models use different text representation techniques. The baseline Logistic Regression model uses TF-IDF, while the DistilBERT model uses a pretrained subword tokenizer to generate numerical representations suitable for transformer-based processing.

```
from sklearn.feature_extraction.text import TfidfVectorizer import pandas as pd

sample_comments = [ "you are a kind person", "you are an insulting person", "you are not a kind person" ]

vectorizer = TfidfVectorizer()

tfidf_matrix = vectorizer.fit_transform(sample_comments)

tfidf_df = pd.DataFrame( tfidf_matrix.toarray(), columns=vectorizer.get_feature_names_out() )

display(tfidf_df.round(3))
```

3.8 Summary of Data Exploration and Key Findings

The exploratory data analysis provided an understanding of the structure, quality, and characteristics of the Jigsaw Toxic Comment Classification dataset.

The analysis produced several important findings:

1. Dataset size: The dataset contains 159,571 comments and eight columns, including six toxicity-related classification labels.

2. Data quality: No missing values, duplicate rows, duplicate comment texts, or empty comments were identified during the initial assessment.
3. Label distribution: General toxicity was the most frequently assigned category, with 15,294 comments, while threats were the least frequent, with 478 comments.
4. Class imbalance: Approximately 90.42% of comments were labeled non-toxic, compared with 9.58% labeled toxic. This imbalance was considered when selecting the baseline classification approach.
5. Multiple labels: Some comments received multiple toxicity labels, demonstrating that different forms of harmful language can overlap.
6. Text characteristics: Comment lengths varied considerably, highlighting the importance of examining text length and understanding input limitations when working with transformer-based models.
7. Text representation: The initial exploration of tokenization and TF-IDF demonstrated how written comments can be transformed into numerical representations for machine learning.

Research Implications

These findings influenced the subsequent development of Maganor AI in several ways.

First, the class imbalance motivated the use of balanced class weights in the Logistic Regression baseline. Second, the overlapping toxicity categories highlighted the complexity of harmful language classification. Third, the variation in comment length and the limitations of traditional text representations provided reasons to investigate a transformer-based model such as DistilBERT.

The exploratory analysis also established the importance of evaluating more than accuracy alone. Precision, recall, F1-score, and error analysis are necessary to understand how well a model identifies harmful comments and where its predictions may be unreliable.

Overall, this stage established the foundation for comparing traditional machine learning with transformer-based language modeling.

4. Baseline Model Development

4.1 Research Methodology and Experimental Design

The first stage of Maganor AI involved developing a traditional machine-learning model to establish a baseline for harmful language detection. This baseline provides a point of comparison for evaluating the performance of a more advanced transformer-based model, DistilBERT.

The experiment was designed as a binary text-classification task. The model receives an online comment as input and predicts whether it belongs to the toxic or non-toxic category.

The experiment followed these main stages:

1. **Data preparation:** The original comments and their toxicity labels were selected from the Jigsaw dataset.
2. **Data splitting:** The dataset was divided into training and testing subsets to evaluate the model on comments it had not previously encountered.
3. **Text representation:** TF-IDF was used to convert written comments into numerical features that a machine-learning algorithm could process.
4. **Model training:** Logistic Regression was trained using the extracted text features.
5. **Model evaluation:** Accuracy, precision, recall, F1-score, and a confusion matrix were used to assess performance.
6. **Error analysis:** False positives and false negatives were examined to understand situations in which the model made incorrect predictions.

This approach establishes a measurable baseline and provides an opportunity to investigate the limitations of traditional text-classification methods before introducing a transformer-based model.

4.2 Defining the Classification Target

The Jigsaw dataset contains six toxicity-related labels: toxic, severe toxic, obscene, threat, insult, and identity hate.

For the initial experiment, the `toxic` column was selected as the target variable. This decision simplified the task into binary classification, allowing the model to distinguish between comments labeled toxic and comments not labeled toxic.

The target variable was defined as follows:

- **0:** Non-toxic comment according to the dataset's `toxic` label.
- **1:** Toxic comment according to the dataset's `toxic` label.

It is important to clarify that this experiment does not independently identify every possible form of online harassment. It learns to predict the dataset's toxic label. The other five labels were retained in the dataset for further analysis.

4.3 Training and Testing Data

The dataset was divided into two subsets:

- **Training set:** 80% of the data, containing 127,656 comments.
- **Testing set:** 20% of the data, containing 31,915 comments.

A stratified train-test split was used to preserve approximately the same proportion of toxic and non-toxic comments in both subsets.

The experiment used a random state of 42 to make the split reproducible. This means that the same observations can be assigned to the training and testing sets when the experiment is repeated using the same settings.

The testing set was kept separate during model training so that it could provide an independent evaluation of the model's predictions.

4.4 TF-IDF Feature Extraction

Machine-learning algorithms generally require numerical input rather than raw text. Therefore, the comments were converted into numerical representations using Term Frequency–Inverse Document Frequency (TF-IDF).

TF-IDF measures how important a word or phrase is within a document relative to the collection of documents.

It considers two main factors:

- **Term Frequency (TF):** How frequently a word or phrase appears in a particular comment.
- **Inverse Document Frequency (IDF):** How informative a word or phrase is across the entire collection of comments. Words appearing in many comments generally receive lower importance.

For this experiment, the `TfidfVectorizer` from Scikit-learn was configured with the following parameters:

Parameter	Setting	Purpose
`max_features`	50,000	Limits the vocabulary to 50,000 features.
`ngram_range`	(1, 2)	Considers individual words and two-word combinations.
`min_df`	2	Ignores terms appearing in fewer than two documents.
`max_df`	0.95	Ignores terms appearing in more than 95% of documents.
`sublinear_tf`	True	Applies logarithmic scaling to term frequency.

The vectorizer was fitted on the training comments only. The same fitted vectorizer was then used to transform the testing comments. This prevents information from the test set from influencing vocabulary construction.

The resulting numerical representations had the following dimensions:

- Training data: 127,656 comments × 50,000 features.
- Testing data: 31,915 comments × 50,000 features.

Using both individual words and two-word combinations allows the model to consider some short expressions rather than treating every word independently. However, TF-IDF does not fully understand sentence meaning, context, sarcasm, or the relationships between words.

4.6 Logistic Regression Model Training

After converting the comments into numerical features using TF-IDF, Logistic Regression was selected as the baseline classification algorithm.

Despite its name, Logistic Regression is commonly used for classification rather than predicting continuous numerical values.

In this experiment, the model learned patterns associated with the toxic and non-toxic categories from the training data. It then used these learned patterns to estimate the probability that an unseen comment belonged to the toxic category.

The model was configured with three important settings:

- **max_iter=1000:** Allows the optimization algorithm up to 1,000 iterations to find suitable model parameters.
- **class_weight="balanced":** Adjusts the importance of each class based on its frequency in the training data. This helps address the imbalance between toxic and non-toxic comments.
- **random_state=42:** Supports reproducibility.

Class balancing was particularly important because toxic comments represented only approximately 9.58% of the dataset, while non-toxic comments represented approximately 90.42%.

Without considering this imbalance, a model could achieve seemingly high accuracy by favoring the majority class while failing to identify many toxic comments.

The trained Logistic Regression model served as the first benchmark for evaluating Maganor AI's ability to classify harmful language.

4.8 Baseline Model Evaluation

The trained model was evaluated using the testing dataset, which contained 31,915 comments that were not used during training.

Five evaluation measures were considered:

1. **Accuracy:** The proportion of all predictions that were correct.
2. **Precision:** The proportion of comments predicted as toxic that were actually labeled toxic.
3. **Recall:** The proportion of dataset-labeled toxic comments that the model successfully identified.
4. **F1-score:** The harmonic mean of precision and recall, providing a balance between the two measures.
5. **Confusion matrix:** A summary of correct and incorrect predictions across both classes.

The model achieved the following results:

Evaluation metric	Result
Accuracy	94.04%
Precision	64.07%
Recall	86.04%
F1-score	73.45%

The model achieved relatively high recall, meaning it identified a large proportion of comments labeled toxic in the testing dataset. However, its precision was lower, indicating that some comments predicted as toxic were not labeled toxic in the dataset.

These results demonstrate why accuracy alone is insufficient for evaluating a classifier trained on an imbalanced dataset.

4.9 Confusion Matrix and Prediction Errors

The confusion matrix provides a more detailed view of the model's classification performance.

The results were:

	Predicted non-toxic	Predicted toxic
Actually non-toxic	27,380	1,476
Actually toxic	427	2,632

The four outcomes are interpreted as follows:

- **True Negatives (27,380):** Non-toxic comments correctly classified as non-toxic.
- **False Positives (1,476):** Comments labeled non-toxic in the dataset but incorrectly predicted as toxic.
- **False Negatives (427):** Comments labeled toxic in the dataset but incorrectly predicted as non-toxic.
- **True Positives (2,632):** Toxic comments correctly classified as toxic.

The confusion matrix reveals two important challenges: false alarms and missed toxic comments. Both are important when considering the reliability and potential real-world use of automated content moderation systems.

4.10 Error Analysis

Although the baseline Logistic Regression model achieved an accuracy of 94.04%, its predictions were not always correct. To understand these limitations, false positives and false negatives were examined separately.

False Positives

The model produced 1,476 false positives. These were comments that the dataset labeled as non-toxic but the model classified as toxic.

Some examples contained offensive expressions, insults, or aggressive language, even though the dataset's primary toxic label was zero. This raises an important research question: does the model sometimes detect harmful expressions that are not reflected in the dataset's annotations?

However, these examples cannot automatically be considered mislabeled. Human review would be needed to determine whether the model or the dataset annotation was more appropriate in each case.

False Negatives

The model produced 427 false negatives. These were comments labeled toxic in the dataset but classified as non-toxic by the model.

Some examples contained insults, profanity, and hostile expressions. Others received relatively low toxicity probabilities despite being labeled toxic.

These errors demonstrate that identifying harmful language is not simply a matter of recognizing individual offensive words. Expressions, context, sentence structure, and the way language is used can influence classification.

Research significance

The error analysis highlights two important limitations:

1. A model may flag comments that are not labeled toxic in the dataset.
2. A model may fail to identify comments that are labeled toxic, even when they contain hostile language.

These observations motivate further investigation into contextual language understanding and the potential advantages of transformer-based models such as DistilBERT.

4.11 Summary of Baseline Findings

The TF-IDF and Logistic Regression baseline achieved an accuracy of 94.04% and an F1-score of 73.45%.

The model demonstrated relatively strong recall (86.04%), suggesting that it could identify most comments labeled toxic in the testing dataset. However, its precision (64.07%) showed that a considerable number of its toxic predictions were not supported by the dataset's primary toxic labels.

The error analysis also revealed that some comments were difficult to classify correctly, particularly when harmful language involved expressions that the model did not recognize as strongly associated with toxicity.

Overall, the baseline experiment established a useful performance benchmark. It also demonstrated the limitations of traditional TF-IDF-based text representation, motivating the next experiment with DistilBERT, a transformer-based language model designed to capture contextual relationships between words.

The baseline results will be used as a reference point when evaluating the next model.

4.12 Limitations of the Baseline Model

Despite its useful performance, the TF-IDF and Logistic Regression model has several limitations.

1. Limited contextual understanding

TF-IDF represents text using word and phrase importance. It does not fully understand the meaning of a sentence or how words relate to one another.

2. Difficulty interpreting sarcasm and indirect language

A comment may communicate hostility without using obvious offensive vocabulary. Similarly, sarcasm can change the intended meaning of an expression.

3. Dependence on vocabulary patterns

The model learns statistical associations between words and toxicity labels. It may struggle with unfamiliar expressions, coded language, or wording that differs from its training examples.

4. Dataset annotation limitations

The model learns from labels assigned in the Jigsaw dataset. These annotations may not capture every form of harmful language or reflect how all communities interpret particular expressions.

5. Class imbalance

Although class weighting was applied, the dataset still contains substantially more non-toxic comments than toxic comments. This imbalance can influence classification performance.

These limitations do not mean that the baseline model is unsuccessful. Rather, they identify areas where further experimentation is necessary.

The next stage introduces DistilBERT to investigate whether contextual language representations can improve harmful language classification.

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5. DistilBERT Model Development

5.1 Introduction to Transformer-Based Models

Traditional machine-learning approaches, such as TF-IDF combined with Logistic Regression, represent text using statistical relationships between words and documents. Although this approach can identify useful vocabulary patterns, it has limitations when interpreting context and relationships between words.

Transformer-based language models offer a different approach. They use attention mechanisms to learn relationships between words within a sequence, allowing them to develop contextual representations of language.

For example, the meaning of a word may change depending on the surrounding sentence. A contextual language model can use information from the surrounding words to produce a more meaningful representation.

In this project, DistilBERT was selected as the second model to investigate whether contextual language representations could improve the classification of toxic comments.

The experiment will compare its performance against the TF-IDF and Logistic Regression baseline using the same testing dataset.

5.2 Why DistilBERT?

DistilBERT is a smaller, more computationally efficient version of BERT (Bidirectional Encoder Representations from Transformers).

It was developed using knowledge distillation, a process in which a smaller model learns to reproduce important behaviors of a larger model.

DistilBERT was selected for this project for several reasons:

1. **Contextual understanding:** It creates contextual representations rather than relying only on individual word frequencies.
2. **Computational efficiency:** It is smaller and generally faster than the original BERT model.
3. **Pretrained language knowledge:** It starts with knowledge learned from large-scale language pretraining, reducing the need to train a language model from scratch.
4. **Practical experimentation:** It provides an opportunity to investigate transformer-based classification within limited computing resources.

The project will use the pretrained `distilbert-base-uncased` model and fine-tune it on the Jigsaw toxicity dataset.

However, DistilBERT is not guaranteed to understand every form of sarcasm, indirect hostility, or culturally specific language. Its actual strengths and limitations will be examined through evaluation and error analysis.

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### 5.5 DistilBERT Tokenization

Tokenization is the process of converting written text into smaller units called tokens so that a language model can process it.

Unlike the TF-IDF approach, which represents text using word and phrase importance, DistilBERT uses a tokenizer designed for its

DistilBERT uses WordPiece tokenization. This means that a word can be represented as one token or divided into smaller subword u

For example, a familiar word may be represented as a single token, while an unfamiliar or complex word may be divided into multi

After tokenization, the text is converted into numerical token IDs. These IDs are then used by the model to generate contextual

Tokenization also introduces special tokens and attention masks, which help the model identify the structure of its input and di
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5.5.5 Maximum Sequence Length

Transformer models process input sequences using a predefined maximum length. For this experiment, the maximum sequence length was set to 128 tokens.

Comments longer than this limit will be truncated, while shorter comments will be padded when necessary to create consistent input dimensions.

This setting was selected as a practical starting point to balance computational efficiency and the amount of text available to the model.

An important limitation is that information appearing after the truncation point may be lost. Therefore, some context in longer comments may not be considered during classification.

5.6 Tokenizing the Training and Testing Data 5.6.1 Understanding padding and truncation

5.6 Preparing the Dataset for DistilBERT

After exploring tokenization using a sample comment, the next step is to convert the training and testing comments into model-ready numerical inputs.

Two important operations are used during this process:

- **Padding:** Adds special padding tokens to shorter comments so that sequences have consistent lengths.
- **Truncation:** Removes tokens beyond the selected maximum sequence length.

For this experiment, the maximum sequence length is 128 tokens. Both padding and truncation are applied to ensure that the inputs have consistent dimensions.

The tokenizer also generates attention masks, which help the model distinguish actual text tokens from padding tokens.

These processed inputs will be used to fine-tune DistilBERT for binary toxicity classification.

5.6.5 Important research note

5.6.5 Tokenization Considerations

The tokenizer transforms each comment into input IDs and attention masks. This allows DistilBERT to process text using numerical representations rather than raw strings.

The maximum sequence length of 128 tokens reduces computational requirements, but it also creates a limitation: longer comments may lose information when truncated.

This is particularly relevant to online harassment research because harmful expressions may appear later in a comment rather than at the beginning. The effect of truncation should therefore be considered when interpreting model errors.

Converting the Data into PyTorch Datasets DistilBERT uses PyTorch tensors to process training examples. Our tokenized data is currently stored in Python lists, so we need to convert it into a format PyTorch can work with.

5.7 Converting Tokenized Data into PyTorch Datasets

After tokenization, the comments are represented by numerical input IDs and attention masks. The next step is to organize these inputs into PyTorch datasets.

PyTorch is a deep-learning framework that supports tensor operations, neural network development, and model training.

For this experiment, each training example contains three components:

- **Input IDs:** Numerical representations of the comment's tokens.
- **Attention masks:** Identify actual tokens and padding positions.
- **Labels:** The corresponding binary toxicity classifications.

These components are converted into PyTorch tensors and organized into datasets. This allows the model to retrieve individual examples or batches during training and evaluation.

5.7.6 Dataset Preparation Summary

The tokenized training and testing comments were converted into PyTorch datasets. Each example contains input IDs, attention masks, and its corresponding toxicity label.

The training dataset contains 20,000 examples, while the testing dataset contains 31,915 examples.

This preparation enables DistilBERT to process the comments in batches during fine-tuning and evaluation.

5.8 Loading the Pretrained DistilBERT Model

After preparing the training and testing datasets, the next step is to load the pretrained DistilBERT model and adapt it for binary toxicity classification.

A pretrained language model has already learned general language patterns during its initial training. Instead of training a language model from scratch, this experiment uses the existing DistilBERT knowledge as a starting point.

The model is loaded with a sequence-classification head, which enables it to assign an input comment to one of two categories:

- Class 0: Non-toxic.
- Class 1: Toxic.

The classification layer is adapted to the two-class task, while the pretrained DistilBERT layers provide the language representations used for prediction.

The model will then be fine-tuned using the Jigsaw training comments.

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### 5.8.6 Model Configuration

The pretrained `distilbert-base-uncased` checkpoint was loaded using the Hugging Face Transformers library.

A sequence-classification head was configured with two output labels to support binary toxicity classification.

The model contains six transformer layers, twelve attention heads, and a hidden representation size of 768 dimensions.

The pretrained model parameters were retained as the starting point for fine-tuning. The classification head was adapted to the

The model was assigned to the available computing device, allowing training to use GPU acceleration when available.
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5.9 Configuring the DistilBERT Training Process

Fine-tuning involves updating a pretrained model's parameters using examples from a specific task. In this experiment, DistilBERT will be fine-tuned to classify online comments as toxic or non-toxic.

Several training parameters must be selected before the process begins.

Learning rate: Determines how much the model's parameters are adjusted during each optimization step. A learning rate that is too high may cause unstable training, while one that is too low may result in slow learning.

Batch size: Determines how many examples the model processes before updating its parameters. A smaller batch size reduces memory requirements.

Number of epochs: Represents how many times the model passes through the entire training dataset.

Optimizer: The algorithm responsible for updating model parameters to reduce the training loss.

For the initial experiment, a relatively small training configuration will be used to balance computational requirements and model performance.

5.9.2 Training Hyperparameters

The following hyperparameters were selected for the initial fine-tuning experiment:

Parameter	Value	Reason
Pretrained model	DistilBERT	Contextual language representation
Training examples	20,000	Manageable initial experiment
Maximum sequence length	128	Reduces computational requirements
Batch size	8	Limits GPU memory usage
Learning rate	2e-5	Conservative fine-tuning rate
Number of epochs	1	Initial performance experiment
Weight decay	0.01	Helps regularize parameter updates
Optimizer	AdamW	Commonly used for transformer fine-tuning

These values establish an initial experimental configuration. They are not assumed to be optimal; further experiments could investigate how changes in learning rate, batch size, and training duration affect performance.

5.9.6 Handling Class Imbalance During Fine-Tuning

The Jigsaw dataset contains substantially more non-toxic comments than toxic comments. This imbalance is important because a model may become biased toward predicting the majority class.

The Logistic Regression baseline addressed this issue using balanced class weights.

The initial DistilBERT experiment uses the standard classification loss without additional class weighting. Therefore, the difference in imbalance-handling strategies will be considered when interpreting the comparison between the two models.

Future experiments could introduce weighted loss functions or other sampling strategies to investigate whether minority-class performance improves.

5.10 Fine-Tuning DistilBERT

Fine-tuning is the process of adapting a pretrained language model to perform a specific task using a labeled dataset.

In this experiment, DistilBERT begins with language representations learned during its original pretraining. It is then trained using the Jigsaw dataset to identify patterns associated with toxic and non-toxic comments.

During training, the model processes batches of comments, generates predictions, calculates the difference between its predictions and the actual labels, and updates its parameters to reduce classification error.

The objective is to adapt the pretrained model to binary toxicity classification while preserving useful language representations learned during pretraining.

The initial experiment uses 20,000 training comments, a batch size of 8, and one training epoch.

5.10.4 Fine-Tuning Results

DistilBERT was fine-tuned using a sample of 20,000 labeled comments from the Jigsaw training dataset.

The experiment used a batch size of 8, a learning rate of 2e-5, and one training epoch.

The training process updated the model parameters to adapt its pretrained language representations to binary toxicity classification.

The training loss and execution time were recorded to document the experiment's computational performance.

These training results provide information about the learning process, while the model's actual classification performance will be assessed separately using the held-out testing dataset.

5.11 DistilBERT Model Evaluation

After fine-tuning, the DistilBERT model was evaluated using the held-out test dataset. This dataset contains comments that were not included in the training process.

The purpose of this evaluation is to measure how effectively the fine-tuned model identifies toxic and non-toxic comments on unseen data.

The model generates prediction scores (logits) for each test comment. These scores are converted into predicted class labels, which are then compared with the original labels to assess model performance.

5.11 DistilBERT Model Evaluation

After fine-tuning the pretrained DistilBERT model on 20,000 training comments, the model was evaluated using the same held-out test set of 31,915 comments used in the baseline experiment.

The model achieved an accuracy of 96.68%, precision of 84.95%, recall of 79.37%, and an F1-score of 82.07%.

Compared with TF-IDF and Logistic Regression, DistilBERT improved accuracy, precision, and F1-score. Its precision increased from 64.07% to 84.95%, indicating that its positive predictions were substantially more reliable. However, its recall decreased from 86.04% to 79.37%, meaning that it missed a greater proportion of comments labelled toxic.

These findings suggest that the pretrained language model provided a stronger overall balance between precision and recall in this initial experiment. Nevertheless, the comparison has an important limitation: Logistic Regression was trained on 127,656 examples, whereas DistilBERT was trained on only 20,000 examples. Therefore, the results should not be interpreted as a fully controlled comparison.

Further investigation of false positives and false negatives is necessary to understand the models' different classification behaviours.

5.12 Confusion Matrix Analysis

The DistilBERT model produced 28,426 true negatives, 430 false positives, 631 false negatives, and 2,428 true positives across the 31,915 test comments.

The results reveal an important difference between the two classification approaches. DistilBERT generated substantially fewer false positives than the TF-IDF and Logistic Regression baseline, reducing them from 1,476 to 430. However, its false negatives increased from 427 to 631.

This indicates that DistilBERT was more conservative when assigning the toxic label. Although this improved the reliability of its positive predictions, it also caused the model to overlook more comments that were labelled toxic in the dataset.

These findings demonstrate that high overall accuracy does not necessarily mean that a model identifies every harmful comment. Both false positives and false negatives matter, particularly in automated content moderation, where incorrect decisions can affect users and online discussions.

5.13 Qualitative Analysis of DistilBERT False Negatives

To investigate the limitations of DistilBERT, five examples were selected from the 631 false-negative predictions. These were comments labelled as toxic in the dataset but classified as non-toxic by the model.

The qualitative examination revealed several potentially challenging language patterns.

First, some comments contained multiple ideas, including personal attacks, accusations, and ordinary discussion. This suggests that identifying harmful language within longer passages may be challenging.

Second, certain comments used indirect insults, sarcasm, or expressions whose meanings depended on the surrounding conversation. Such examples demonstrate that harmful meaning cannot always be determined through individual words.

Third, some comments mixed insulting expressions with joking language, making their intended tone ambiguous. Another example contained antisemitic language within a historical quotation, highlighting the importance of distinguishing quoted harmful content from the context and purpose of the surrounding discussion.

These observations suggest that automated toxicity classification involves more than identifying offensive vocabulary. Understanding context, tone, indirect expressions, and quoted material remains challenging.

However, this analysis is exploratory and based on a small selection of false negatives. Further examination of a larger sample and human review would be necessary before drawing broader conclusions.

5.14 Confidence Analysis of False-Negative Predictions

To investigate the confidence of DistilBERT's incorrect predictions, the predicted probability of the toxic class was examined for all 631 false-negative comments.

The average toxic probability was 0.0787 (7.87%), while the median was 0.0221 (2.21%). The probabilities ranged from 0.000543 to 0.492833.

These findings indicate that many false-negative predictions were made with high model confidence rather than occurring only near the classification threshold.

The examination of the most confident mistakes revealed that some comments contained direct insults, aggressive expressions, and extended hostile discussions. Nevertheless, DistilBERT assigned them very low probabilities of belonging to the toxic class.

This observation raises questions about how pretrained language models learn associations between words, context, and toxicity labels. It also demonstrates that high confidence does not necessarily guarantee a correct classification.

However, the probability analysis alone cannot establish why these errors occurred. Further experiments involving contextual examples, model explanations, and human review would be necessary to investigate the underlying causes.

5.15 Investigation of Comment Length and Classification Errors

The relationship between comment length and DistilBERT classification errors was investigated by comparing false-negative comments with correctly identified toxic comments.

The 631 false-negative comments had an average length of 408.31 characters and a median length of 167 characters. In comparison, the 2,428 correctly identified toxic comments had an average length of 268.55 characters and a median length of 111 characters.

The false-negative group therefore contained longer comments on average. This observation suggests a possible association between comment length and classification difficulty.

One potential explanation is input truncation. The DistilBERT tokenizer was configured with a maximum sequence length of 128 tokens. Consequently, longer comments could lose information beyond the model's input limit. If important expressions appeared in the truncated portion, the model would not be able to use them during classification.

Another possible explanation is that longer comments contain a mixture of ordinary discussion and harmful expressions, making their overall meaning more difficult to interpret.

These findings do not establish that comment length caused the classification errors. Further experiments using different sequence lengths would be required to investigate the potential influence of truncation.

5.16 Relationship Between Input Length and False-Negative Rates

To investigate whether input length was associated with classification errors, all 3,059 toxic comments in the test set were divided into two groups according to the tokenizer's maximum sequence length of 128 tokens.

Among the 2,675 comments containing 128 tokens or fewer, 500 were incorrectly classified as non-toxic, resulting in a false-negative rate of 18.69%.

Among the 384 comments exceeding 128 tokens, 131 were incorrectly classified as non-toxic, resulting in a false-negative rate of 34.11%.

The false-negative rate for longer comments was approximately 1.83 times that of shorter comments, representing a difference of 15.42 percentage points.

These findings indicate an association between longer input sequences and a higher rate of missed toxic comments in this experiment. One possible contributing factor is tokenizer truncation, which may remove relevant information from longer comments.

However, this analysis does not establish truncation as the direct cause of the increased error rate. Other linguistic and contextual factors may also influence classification performance. A controlled experiment using a longer maximum sequence length would help investigate this possibility.

5.17 Statistical Analysis of Comment Length and Classification Errors

A chi-square test of independence was conducted to investigate the relationship between comment length and DistilBERT classification outcomes.

The test produced a chi-square statistic of 47.8484, with one degree of freedom and a p-value of approximately 4.60×10^{-12}.

Since the p-value was substantially below the significance threshold of 0.05, the null hypothesis of independence was rejected.

The results provide strong statistical evidence of an association between comment length group and classification outcome. Toxic comments exceeding 128 tokens had a false-negative rate of 34.11%, compared with 18.69% for comments containing 128 tokens or fewer.

This finding supports further investigation into the effect of input sequence length on toxicity classification. However, statistical association does not establish causation. In particular, the experiment does not prove that tokenizer truncation alone caused the higher error rate.

A controlled experiment using a longer input sequence length is therefore proposed to investigate whether retaining more textual information can improve classification performance.

5.18 Second DistilBERT Experiment: Increasing the Input Length to 256 Tokens

To investigate whether providing additional textual context could improve toxic comment classification, a second DistilBERT experiment was conducted using a maximum sequence length of 256 tokens.

The experiment used the same pretrained model architecture (`distilbert-base-uncased`), 20,000 training examples, and 31,915 test examples. The training configuration remained unchanged, including one training epoch, a batch size of 8, a learning rate of 2e-5, and a weight decay of 0.01.

The main experimental adjustment was increasing the maximum input length from 128 to 256 tokens.

Evaluation Results

Metric	128 Tokens	256 Tokens
Accuracy	96.68%	96.64%
Precision	84.95%	84.76%

Metric	128 Tokens	256 Tokens
Recall	79.37%	79.24%
F1-score	82.07%	81.91%

The 256-token experiment did not demonstrate an improvement in the overall evaluation metrics. Its F1-score was 81.91%, compared with 82.07% for the 128-token experiment.

The confusion matrix for the 256-token model contained 28,420 true negatives, 436 false positives, 635 false negatives, and 2,424 true positives.

Comment Length and Classification Errors

A further investigation examined whether comment length was associated with false-negative predictions.

Comment Length	Total Toxic Comments	Missed Comments	False-Negative Rate
256 tokens or fewer	2,915	586	20.10%
Over 256 tokens	144	49	34.03%

The chi-square test produced a statistic of 15.34 with a p-value of 0.0000898. Cramér's V was 0.0708, indicating a small association between comment-length group and false-negative classification.

These findings suggest that longer comments remained more challenging for the model. However, the analysis does not establish that truncation alone caused the classification errors.

Conclusion

Increasing the maximum input length from 128 to 256 tokens did not improve the model's overall classification performance in this experiment. Although the longer input limit allowed more textual information to be processed, the model continued to miss a substantial proportion of toxic comments.

This observation highlights an important limitation of automated language classification: providing additional text does not necessarily guarantee better contextual understanding. The findings also suggest that detecting harmful language requires attention to more than input length, including linguistic context, indirect hostility, and ambiguous expressions.

Chapter 6: Discussion of Findings

6.1 Comparison of the Three Classification Models

The experiments demonstrated differences in how traditional machine learning and transformer-based models classified toxic online comments.

The TF-IDF and Logistic Regression baseline achieved an accuracy of 94.04%, precision of 64.07%, recall of 86.04%, and F1-score of 73.45%.

The first DistilBERT experiment, using a maximum input length of 128 tokens, achieved an accuracy of 96.68%, precision of 84.95%, recall of 79.37%, and F1-score of 82.07%.

The second DistilBERT experiment, using 256 tokens, achieved an accuracy of 96.64%, precision of 84.76%, recall of 79.24%, and F1-score of 81.91%.

These results indicate that DistilBERT achieved higher precision and F1-score than the baseline in the experiments conducted. However, Logistic Regression achieved higher recall, identifying a larger proportion of the toxic comments.

This difference illustrates that model performance depends on the evaluation metric being considered. A model that produces fewer false alarms does not necessarily identify more of the harmful comments in a dataset.

An important methodological consideration is that the baseline model was trained on 127,656 examples, whereas each DistilBERT experiment used 20,000 training examples. Consequently, the results represent the performance of the specific experimental configurations rather than a perfectly controlled comparison of model architectures.

6.2 The Effect of Increasing Input Length

One objective of the second DistilBERT experiment was to investigate whether increasing the maximum input length could improve toxic comment classification.

The results did not demonstrate an improvement. The F1-score decreased slightly from 82.07% at 128 tokens to 81.91% at 256 tokens.

The difference was small, and the results should not be interpreted as proof that shorter inputs are generally better. Nevertheless, the experiment demonstrated that providing a model with more text does not automatically guarantee improved classification performance.

This observation raises further questions about how language models process context and whether additional information is always useful for identifying harmful expressions.

6.3 Comment Length and False-Negative Predictions

The error analysis revealed an association between comment length and false-negative predictions in both DistilBERT experiments.

In the 128-token experiment, comments exceeding 128 tokens had a false-negative rate of 34.11%, compared with 18.69% for shorter comments.

In the 256-token experiment, comments exceeding 256 tokens had a false-negative rate of 34.03%, compared with 20.10% for shorter comments.

Chi-square tests indicated statistically significant associations in both experiments. However, the corresponding Cramér's V values (0.125 and 0.071) indicated small associations.

These findings suggest that longer comments remain challenging for the models. Truncation may contribute to this difficulty because portions of longer comments are not processed, but other factors, including linguistic complexity and contextual ambiguity, may also influence classification errors.

The findings establish an association rather than a causal relationship.

6.4 Understanding False Positives and False Negatives

The qualitative error analysis revealed two important limitations of automated toxic language classification.

False positives occurred when comments were predicted as toxic despite having a non-toxic primary label in the dataset. Some of these comments contained potentially offensive expressions or secondary toxicity labels, demonstrating that annotation decisions and the meaning of individual words can be complex.

False negatives occurred when comments labelled as toxic were classified as non-toxic. Some examples contained personal attacks, indirect insults, sarcastic expressions, or hostile language embedded in longer discussions.

These observations suggest that identifying harmful language is not simply a matter of detecting individual offensive words. The surrounding discussion, intended meaning, and linguistic context can influence how an expression should be interpreted.

However, individual examples cannot establish that a model consistently fails on a particular linguistic category.

6.5 Implications for Automated Content Moderation

The experiments demonstrate both the potential and limitations of automated language classification.

DistilBERT achieved stronger precision and F1-score than the TF-IDF baseline, but it still failed to identify some comments labelled as toxic. Meanwhile, the baseline achieved higher recall but generated more false-positive predictions.

In a practical moderation environment, these trade-offs matter. False positives can incorrectly flag legitimate discussions, while false negatives can allow harmful content to remain undetected.

Therefore, automated classification should be treated as a decision-support mechanism rather than a complete replacement for human judgment.

Human review remains important, particularly when comments contain sarcasm, indirect hostility, ambiguous expressions, or culturally specific language.

Chapter 7: Research Limitations and Responsible AI Considerations

7.1 Dataset Limitations

This research used the Jigsaw Toxic Comment Classification dataset, which contains English-language comments from Wikipedia discussions.

Although the dataset provides useful examples for studying toxic language classification, it does not represent all forms of online communication.

In particular, the dataset does not adequately represent Ethiopian social-media environments, local cultural expressions, or languages such as Amharic and Afaan Oromo.

Consequently, the models developed in this project cannot be assumed to identify harmful language accurately across Ethiopian online communities.

The dataset also contains human-assigned labels, which may not capture every interpretation of a comment. Some expressions may be ambiguous, context-dependent, or interpreted differently by different annotators.

7.2 Model and Experimental Limitations

The experiments were conducted under limited computational and time resources.

The Logistic Regression model used 127,656 training examples, while each DistilBERT experiment used a randomly selected subset of 20,000 examples. This difference limits the fairness of direct architectural comparisons.

Additionally, each DistilBERT configuration was trained for only one epoch. The experiments were not repeated across multiple random seeds, and no hyperparameter search was conducted.

Therefore, the observed differences between the 128-token and 256-token experiments should be interpreted as findings from individual experimental runs rather than definitive evidence that one sequence length is universally better.

The evaluation focused on binary toxic versus non-toxic classification using the primary toxic label. Although the dataset contains six toxicity-related labels, the project did not develop a complete multi-label classification system.

7.3 Limitations of Contextual Understanding

The error analysis identified false-positive and false-negative predictions involving potentially ambiguous or context-dependent language.

However, the project did not independently establish the intended meaning of every comment or conduct a formal human annotation study.

The observed errors therefore provide examples of classification limitations, not proof that the models cannot understand sarcasm, indirect hostility, or gender-based harassment in every situation.

Further investigation would require carefully selected examples, additional human evaluation, and datasets representing different linguistic and cultural contexts.

7.4 Ethical Considerations

Automated toxicity classification can influence how people communicate and participate in online spaces.

Incorrectly classifying legitimate comments as harmful may discourage participation, while failing to identify genuinely harmful expressions may leave affected individuals without adequate protection.

There is also a risk of bias when a model is trained on data that does not adequately represent the communities in which it may eventually be used.

For these reasons, the results of this project should not be used as the sole basis for deleting content, restricting accounts, or making judgments about individuals.

Any future practical application should include appropriate human oversight, transparent explanations of system limitations, and opportunities to challenge incorrect classifications.

7.5 Privacy and Responsible Data Use

The research uses an existing public research dataset rather than collecting private conversations or personal messages.

The project focuses on classifying textual content, not identifying individuals or inferring their personal characteristics.

Any future extension involving social-media data collection should consider informed consent where appropriate, data minimization, secure storage, and protection against exposing sensitive personal information.

7.6 Future Research Directions

Several areas could strengthen this research in the future:

- Investigating how sarcasm, indirect hostility, and context influence toxic language classification.
- Evaluating models on carefully curated Ethiopian social-media examples, with appropriate privacy safeguards.
- Exploring multilingual toxicity classification for Amharic and Afaan Oromo.
- Conducting human evaluation to examine ambiguous predictions and annotation disagreements.
- Testing model performance across different training configurations and random seeds.
- Investigating methods for improving the detection of long and context-dependent comments.

These directions represent opportunities for further research rather than capabilities already implemented in the current prototype.

Chapter 8: Conclusion

8.1 Conclusion

This research explored the challenges of identifying toxic and harmful language in online comments using traditional machine learning and transformer-based natural language processing.

The study began with an exploration of the Jigsaw Toxic Comment Classification dataset, followed by the development of a TF-IDF and Logistic Regression baseline model. Two additional experiments were conducted using DistilBERT, with maximum input lengths of 128 and 256 tokens.

The results demonstrated that the DistilBERT model using 128 tokens achieved the highest F1-score of 82.07%, while the Logistic Regression baseline achieved the highest recall of 86.04%. The 256-token experiment produced an F1-score of 81.91%, showing no measurable improvement over the shorter input configuration in this particular experimental setup.

Further investigation of false-positive and false-negative predictions revealed that both models had difficulty with certain comments. Statistical analysis also identified an association between longer comment length and false-negative predictions in both DistilBERT experiments.

These findings highlight an important challenge in natural language processing: harmful language cannot always be identified through individual words or surface-level patterns. The surrounding context, structure, and intended meaning of an expression can influence its interpretation.

The research also demonstrated that improving one evaluation metric does not necessarily improve all aspects of classification performance. Precision, recall, and false-negative rates must be considered together when evaluating systems intended to identify harmful language.

Although the project provides an initial exploration of automated toxic language classification, its findings are limited by the dataset, computational resources, and experimental design. The results should not be generalized to Ethiopian social-media environments or languages without further evaluation.

Ultimately, this project strengthened my understanding of both the capabilities and limitations of language models. It also raised broader questions about how artificial intelligence systems interpret human communication and how they can be developed and evaluated more responsibly.

8.2 Final Reflection

The most important outcome of this research was not simply achieving a particular model score, but discovering how much complexity exists behind seemingly straightforward language classification.

The experiments showed that a model can produce strong overall metrics while still misunderstanding individual comments. Investigating these errors shifted the focus of the project from simply building a classifier toward understanding why automated systems sometimes make incorrect decisions.

This experience reinforced the importance of critical evaluation, contextual understanding, and responsible